

Xin Meng 孟昕

Medical AI Researcher | 3D Medical Imaging · Multimodal AI · Trustworthy AI

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Research Profile

I study reliable medical AI beyond headline benchmark scores, focusing on **3D medical imaging**, **multimodal image-text modeling**, and **reliability auditing**. I am preparing for **2027 PhD applications** in medical imaging, biomedical engineering, computer science, and related AI fields.

Education

China Medical University, Shenyang, China	2024.09 – 2027.06
<i>Master of Library and Information Science (MLIS); research focus: Medical AI</i>	
Jilin University, Changchun, China	2018.08 – 2022.06
<i>B.Mgmt. in Archival Science; B.Econ. in International Economics and Trade</i>	

Research Focus

- 3D medical imaging and generalization:** lung CT classification, longitudinal imaging, registration, calibration, matched architecture controls, external-cohort validation, and source-free adaptation.
- Multimodal and grounded medical AI:** image-text fusion, clinical-text provenance, guideline-grounded RAG, evidence sufficiency, abstention, and structured decision realization.
- Trustworthy evaluation:** shared failure patterns, distribution shift, test-time update risk, training-data lineage, unsupported claims, selective referral, and reproducible audit pipelines.

Selected Research Outputs

Selected to represent my current PhD research direction. † Equal contribution. Double-anonymous submissions are described without venue identifiers.

Source-calibrated CT deep learning for lung nodule malignancy risk: external validation and matched-control analysis across public cohorts

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Yuhong Zhao, Qing Chang
Manuscript under review – Cancer Imaging

Developed on LUNA25 with source-fitted calibration and fixed thresholds; externally evaluated on LNDb and LIDC-IDRI with matched CNN/Mamba/attention controls.

Dual-teacher distillation and source-free adaptation for 3D lung-nodule classification

Xin Meng, co-first author (listed first)
Manuscript under review

Studies a SegMamba student with CNN/transformer teachers, gradient-level distillation behavior, external LNDb transfer, optional test-time adaptation, and compute auditing.

Mining Multi-Model Failure Patterns for Reliability Auditing of 3D Lung Nodule Malignancy Predictors

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Yuhong Zhao, Qing Chang
Manuscript under review

Mines shared errors, consensus failures, high-confidence label discordance, subgroup enrichment, and selective-referral behavior from seven 3D predictors on LUNA25 and LNDb.

Auditable Decision-Invariant Retrieval-Augmented Lung-RADS Report Realization: A Verified-Subset Technical Evaluation

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Yuhong Zhao, Qing Chang
Manuscript under review – IEEE Journal of Biomedical and Health Informatics

Separates fixed rule decisions from retrieval and language realization, with citation, abstention, leakage, and unsupported-claim audits over lung-nodule evidence.

A Benefit-Risk Audit of Test-Time Update Budgets for Longitudinal Medical Image Registration: A GrowthMamba Case Study

Xin Meng[†], Xiaopeng Li[†], Huixin Yu[†], Qing Chang, Yuhong Zhao
Oral presentation, 32nd Chinese Medical Association Conference on Medical Informatics, 2026

Audits frozen inference versus multiple test-time update budgets using similarity and Jacobian-folding risk on longitudinal CT and cross-domain MRI.

Selected Additional Research

Additional selected work extending the same reliability and grounded-AI themes.

Clinical-text leakage and text-dominance auditing in multimodal lung-nodule assessment. Xin Meng, co-first author (listed first). Manuscript under review.

Source-bounded answerability and over-decision benchmarking for grounded RAG. Xin Meng, co-first author (listed first). Manuscript under review.

General-purpose VLMs versus task-specific 3D models for pulmonary nodule malignancy prediction. Xin Meng, co-first author (listed first). Manuscript under review.

Research Experience

National Health Commission Health Development Research Center 2025.11 – 2026.02
Remote Research Assistant, Prof. Wudong Guo’s group

- Collected and harmonized multi-province expert consensus documents and group standards on off-label drug use; built a structured comparison table for retrieval and downstream analysis.

Technical Skills

Programming	Python, PyTorch, pandas, SQL, Git/GitHub; experiment automation and reproducible analysis pipelines.
Medical imaging	3D CT preprocessing, HU normalization, voxel-spacing handling, longitudinal pairing, patient-level splitting, external-cohort evaluation.
Modeling	3D CNNs, Mamba/state-space models, Transformers, knowledge distillation, multimodal fusion, RAG, agent workflows.
Evaluation	Patient-clustered bootstrap, calibration, fixed operating points, ablations, matched controls, subgroup/failure analysis, selective referral, provenance audits.
Languages	Chinese (native); English: IELTS 6.5 (Reading 7.5, 2024), CET-6 502.

Academic Communication

- Oral presenter and youth-group science-communication presenter at the 32nd Chinese Medical Association Conference on Medical Informatics (2026).

Prior Data Experience

Shanghai Zhizhongda Information Technology Co., Ltd. – Data Analyst, 2024.05–2024.08. SQL/Tableau analysis of large quotation and order tables, competitor-price monitoring, and pricing-rule support.

Gold Hongye Paper Group (APP / Qingfeng) – Data Analysis Management Trainee, 2023.04–2024.05. Campaign and pricing analytics across five regions and 100+ core SKUs, including JD.com 618 campaign support.